



ARRAYS AND MULTISET TYPES IN SQL

TINA FRANCIS



In SQL, arrays and multisets are data types used to store collections of values, but they have distinct characteristics:



Arrays

Definition:

An array is an ordered collection of elements, all of the same data type.





Key Features:

- Elements are stored in a specific sequence.
- Duplicate values are allowed.
- Arrays have a fixed size, defined at the time of creation.



example:

```
CREATE TYPE colors_array AS VARRAY(5) OF VARCHAR(20);  
CREATE TABLE products (  
    id INT,  
    name VARCHAR(50),  
    colors colors_array  
);  
INSERT INTO products (id, name, colors)  
VALUES (1, 'Shirt', colors_array('Red', 'Blue', 'Green'));
```


MULTISETS

Definition:

- A multiset (or bag) is an unordered collection of elements, where duplicates are allowed.

Key Features:

- No specific order for elements.
- Can dynamically grow in size.



example:

```
CREATE TYPE fruits_multiset AS TABLE OF VARCHAR(20);  
CREATE TABLE shopping_cart (  
    id INT,  
    items fruits_multiset  
);  
INSERT INTO shopping_cart (id, items)  
VALUES (1, fruits_multiset('Apple', 'Banana', 'Apple', 'Orange'));
```


